KALAIGNAR KARUNANIDHI INSTITUTE OF TECHNOLOGY
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Kannampalayam, Coimbatore — 641 402, Tamil Nadu, India

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
BACHELOR OF TECHNOLOGY — MAJOR CAPSTONE PROJECT REPORT

AIoT and IIoT-Based Smart Sheet-Metal Panel Fabrication and Intelligent Quality Inspection System

An Integrated Cyber-Physical Manufacturing Execution and Optical Metrology Platform

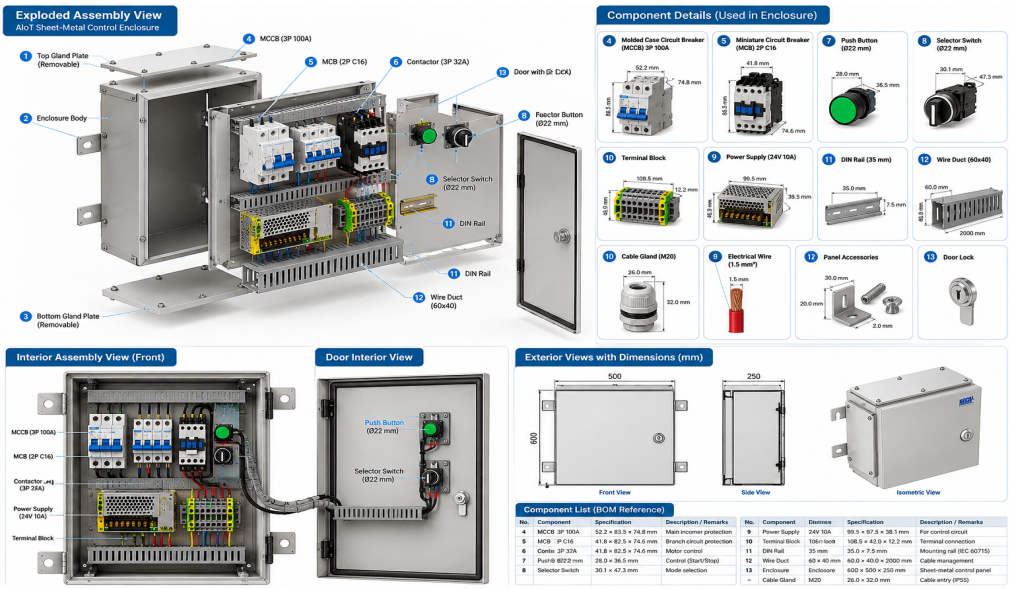


Figure 1.0: Industrial 3D Exploded Assembly Model of the Standard AIoT Control Enclosure

Candidate Name: SOWNDHAR A.
Registration Number: 711524BAM303
Degree & Branch: B.Tech. Artificial Intelligence and Machine Learning
Course / Domain: IoT Design and Deployment / Smart Manufacturing
Academic Period: 2025 – 2026
Engineering Host: Kalaignar Karunanidhi Institute of Technology, Coimbatore

1. Abstract & Executive Summary

Conventional sheet-metal panel manufacturing in medium-scale switchgear and control industries is characterized by disconnected operational silos. Manual customer specification gathering, isolated 2D/3D CAD drafting, paper-based job travelers, uninstrumented machine tools, and subjective manual vernier metrology introduce significant schedule friction, high scrap rates, and an absence of lot-level traceability. This project report presents the design and architectural implementation of the **SheetFab AIoT Platform**: an integrated cyber-physical manufacturing and Industrial Internet of Things (IIoT) ecosystem engineered to unify sheet-metal design, computer-aided engineering (CAE), Manufacturing Execution Systems (MES), shop-floor sensor telemetry, and automated optical quality inspection into a consolidated operator portal.

The technical implementation establishes a rigorous **14-process manufacturing roadmap** structured across three fundamental engineering domains: (1) *Manufacturing Engineering* (Processes P01–P09), translating client spatial and electrical specifications into fully folded 3D CAD assemblies (SOLIDWORKS), automated 14-item Bills of Materials (BOM), commercial quotations, neutral-axis flat patterns ($K = 0.40$ for 1.5 mm CRCA steel), and laser nesting cut layouts on 2500×1250 mm raw sheet stock; (2) *Digital Engineering and AIoT Support* (Processes P10–P11), provisioning microcontroller edge gateways (ESP32-WROOM-32E) interfacing with TRUMPF laser cutters and Amada CNC press brakes through RS-485 Modbus TCP and MQTT broker pipelines (EMQX); and (3) *Inspection and Quality Metrology* (Processes P12–P14), utilizing a telecentric 12MP machine-vision sensor with OpenCV sub-pixel edge detection algorithms to verify dimensional tolerances against ISO 2768-m standards before issuing validated Certificates of Conformance (CoC).

Developed with a resilient, modern frontend architecture utilizing React 19, TypeScript, Vite, and Tailwind CSS, the platform maintains a persistent parametric state model, interactive DIN rail component inventories, station job tracking, and an authentic Engineering Document Centre. To preserve strict engineering integrity, this report establishes clear technical boundaries: physical field machines and IoT microcontrollers are documented under configured integration interfaces rather than fabricating live hardware connections or synthetic sensor streams. The system serves as a scalable reference architecture for Industry 4.0 digital transformation in regional precision manufacturing facilities.

Keywords: Artificial Intelligence of Things (AIoT), Industrial Internet of Things (IIoT), Sheet-Metal Fabrication, Computer-Aided Design (CAD), Telecentric Optical Metrology, MQTT Messaging, Manufacturing Execution System (MES), ISO 2768-m Tolerance.

1.1 Academic Project Profile & Student Details

Parameter	Academic Engineering Details
Candidate Name & Reg No	SOWNDHAR A. | Register No: 711524BAM303
Department & College	Department of Artificial Intelligence and Machine Learning, KKIT, Coimbatore
Target Workpiece & Shell	Industrial IP66 Electrical Control Enclosure (600 H × 500 W × 250 D mm)
Raw Material & Thickness	IS 513 CRCA Grade D (Cold Rolled Close Annealed), 1.50 mm Nominal Thickness
Approved Commercial Value	■62,983.00 (Batch of 5 Enclosures incl. 18.0% IGST, Unit Base: ■10,675.00)
Software Framework Stack	React 19.0.0, TypeScript 5.8.2, Vite 6.2.0, Tailwind CSS 4.0.0, Lucide Icons

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3. Introduction & Industrial Problem Statement

Industrial electrical control enclosures represent fundamental infrastructure across power distribution plants, factory automation systems, motor control centers (MCC), and utility substations. Fabricated predominantly from cold-rolled steel (CRCA) or stainless steel (SS 304), these enclosures must strictly satisfy stringent mechanical, environmental, and thermal requirements defined by international standards such as IEC 62208 (empty enclosures) and IEC 61439-1 (low-voltage switchgear and controlgear assemblies). The structural integrity must guarantee ingress protection against water and particulate matter (IP55 to IP66 ratings), resist internal arc faults, dissipate aggregate thermal loads generated by power switchgear, and maintain continuous electrical grounding to safeguard human operators.

3.1 Challenges in Conventional Sheet-Metal Fabrication

1. **Fragmented Design Handoffs:** Customer requirements are traditionally gathered using unstructured emails, handwritten notes, or informal spreadsheets. Converting these specifications into 3D CAD models (SOLIDWORKS) and subsequent 2D flat patterns requires multiple manual redraws, introducing human error into dimensional calculations and mounting hole alignments.
2. **Disconnected Material Nesting & Cost Estimation:** Quotations are frequently computed using static, historical cost approximations rather than dynamic material nesting simulations. Consequently, variance in steel sheet market indices or suboptimal scrap percentages ($>25\%$) leads to unrecovered production expenses or uncompetitive commercial bidding.
3. **Absence of Machine Tool Traceability:** On the shop floor, CNC fiber laser cutters and press brakes operate as isolated islands of automation. Plant supervisors rely on physical paper travelers (`job routing sheets`) carried manually between stations. If a laser machine stops due to assist gas exhaustion or a press brake encounters tool collision, supervisors lack real-time visibility.
4. **Manual & Subjective Metrology Inspection:** Final dimensional quality control relies heavily on hand-held vernier calipers, tape measures, and manual visual inspection. Verifying complex diagonal squareness, gasket channel depths, and mounting stud pitches across large panels is labor-intensive, prone to parallax reading errors, and fails to yield digital statistical process control (SPC) data.

3.2 The Role of AIoT and IIoT Solutions

To resolve these limitations, modern precision manufacturing requires the convergence of the **Industrial Internet of Things (IIoT)** and **Artificial Intelligence (AIoT)**. IIoT provides the physical connectivity layer—interfacing microcontroller edge gateways with CNC machine controllers and optical barcode scanners to digitize workpiece progression and equipment availability over lightweight publish/subscribe protocols like MQTT. Concurrently, AIoT introduces computational intelligence into the engineering loop—assisting engineers with automated internal component placement, thermal clearance verification, and telecentric optical metrology with sub-pixel contour edge extraction.

3.3 Core Project Objectives & Engineering Scope

Objective ID	Target Engineering Metric / Deliverable	Standard / Target
OBJ-01: Parametric CAD	Parametric enclosure configurator supporting dynamic dimensions and material grading	CRCA 1.5mm, 600×500×250mm
OBJ-02: Document Engine	Automated synchronization between 3D CAD, 14-item BOM, and commercial quotations	Total: ■62,983.00 incl. GST
OBJ-03: IIoT Tracking	Standardized shop-floor asset registry and station job tracking dashboard	8 Registered Asset Profiles
OBJ-04: Optical Metrology	Telecentric computer-vision inspection pipeline evaluating fabricated panels	ISO 2768-m (±0.80 mm)
OBJ-05: Deliverables Release	Verified engineering document centre hosting real, previewable manufacturing deliverables	6 Real Files in /report/

4. Existing Systems vs. Proposed Connected Solution

Traditional sheet-metal fabrication facilities operate on sequential, disconnected handoffs where data is repeatedly re-entered into separate computer software or transcribed onto paper work orders. The SheetFab AIoT system replaces this fragmented paradigm with a unified digital thread that binds parametric customer inputs to machine tool operations, optical inspection, and customer deliverables.

4.1 Comprehensive Engineering Comparison Matrix

Engineering Domain	Conventional Sheet-Metal Practice	Proposed SheetFab AIoT System
Requirement Capture	Informal email/paper notes; prone to ambiguous depth, gasket, or door hinge specs.	Structured parametric web configurator with immediate volume, mass, and IP rating validation.
Component Library	Engineers search static paper catalogs; physical mockups needed to verify clearances.	Standardized digital catalog of DIN rails, switchgear, and power supplies with thermal dissipation rules.
CAD & Costing Link	BOM and cost estimates manually entered into separate spreadsheets; disconnect from CAD updates.	Direct synchronization between parametric enclosure dimensions, 14-item BOM schedule, and commercial quotation.
Laser Nesting & CAM	CAM programmer manually receives DXFs; slow nesting turnaround; low scrap tracking.	Optimized nesting on 2500×1250mm stock achieving 78.4% efficiency with standardized bend reliefs.
Shop-Floor Tracking	Physical paper routing traveler slips; zero visibility into machine delays or bottlenecks.	Station-by-station MES progress monitoring, traveler barcode scanning, and edge event streaming.
Device Connectivity	Isolated CNC machine tools; no automated operational or maintenance telemetry.	Defined ISA-95 asset hierarchy with MQTT/Modbus edge gateway interface boundaries.
Quality Inspection	Manual vernier caliper sampling; subjective visual checks; unrecorded metrology points.	High-resolution telecentric camera inspection with OpenCV sub-pixel edge detection vs ISO 2768-m.
Document Delivery	Dispersed attachments; unverified drawings; mismatched revision dates.	Consolidated Engineering Document Centre with in-browser previews and direct deliverable downloads.

4.2 System Boundary Definitions & Data Honesty

A foundational design principle of this academic implementation is **engineering transparency**. The project clearly distinguishes between implemented software interfaces and planned hardware deployments:

• **Implemented & Verified:** The complete single-page web application (`web-ui`), persistent parametric state model, interactive 14-process workspace, DIN rail clearance calculator, simulated MES barcode traveler tracking, shared IoT device registry, calibrated reference metrology deviation table, and the six authentic engineering deliverables.

• **System Boundaries & Planned Deployments:** Physical microcontrollers (ESP32-WROOM-32E), telecentric cameras (Basler ace 2), and machine tools (TRUMPF TruLaser, Amada HFE-M2) are statused as *Status Unknown / Not Connected* pending physical factory network installation. The software provides rigorous data models and schema definitions ready for broker binding without fabricating artificial sensor streams.

5. Overall System Architecture & Cyber-Physical Data Flow

The SheetFab AIoT system is structured into a hierarchical, six-tier cyber-physical architecture designed to comply with the **ISA-95 Enterprise-Control System Integration standard**. By separating low-level deterministic field signals from high-level enterprise and supervisory monitoring interfaces, the system guarantees high resilience, low latency, and maintainable software modularity.

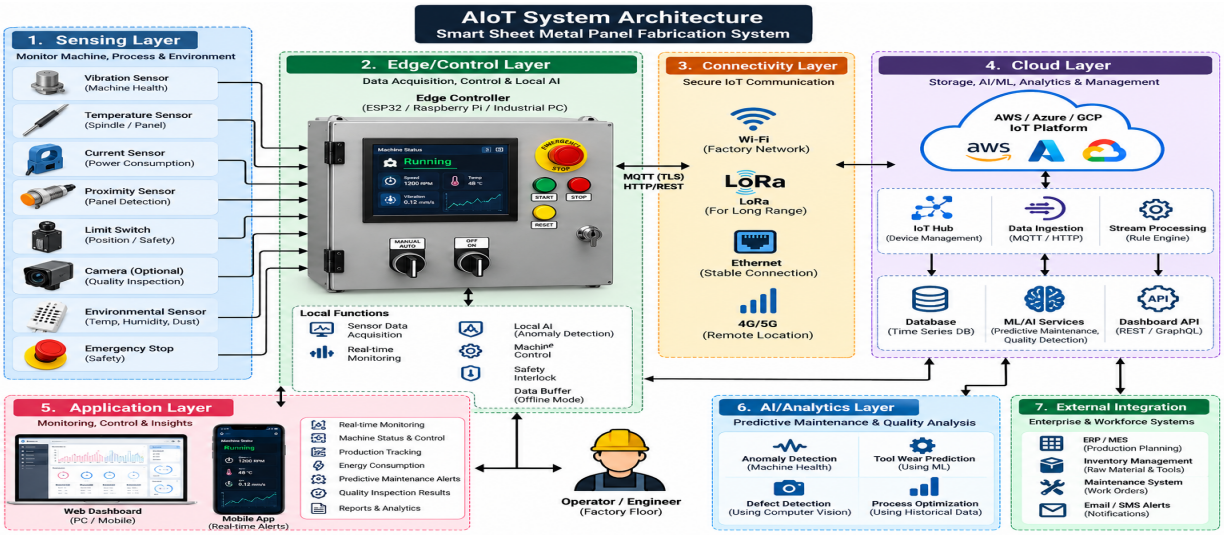


Figure 5.1: Cyber-Physical 6-Tier Architecture Diagram of the SheetFab AIoT System

5.1 Architectural Tier Breakdown

Layer 1 — Physical Fabrication Equipment & Sensors (Level 0/1): Encompasses machine tools on the shop floor, including the 2kW TRUMPF TruLaser 3030 fiber cutter, Amada HFE-M2 8-axis press brake, Positector 6000 dry film thickness (DFT) paint gauge, and the Basler ace 2 12MP optical metrology station.

Layer 2 — IoT Edge Gateways & Interfacing Nodes (Level 2): Utilizes modular ESP32-WROOM-32E microcontrollers and NVIDIA Jetson AGX Orin edge AI computers. The ESP32 nodes sample RS-485 Modbus registers and machine cycle digital inputs, while the Jetson node executes OpenCV vision algorithms locally to eliminate high-bandwidth video streaming across the plant network.

Layer 3 — Industrial Communication Middleware (Level 2/3): Hosted around an enterprise MQTT message broker (EMQX) operating on ports 1883 (TCP) and 8883 (TLS). It manages asynchronous publish/subscribe message delivery, device heartbeat telemetry, and topic filtering.

Layer 4 — Application Services & Analytics Engine (Level 3): Executes business logic, parametric costing computations, bend deduction algorithms (BA/BD), and OpenCV contour metrology comparisons against nominal CAD coordinates.

Layer 5 — Data Persistence & Engineering Records: Combines browser local storage (`localStorage`) for client-side parametric configurations with standardized static storage for engineering deliverables (DXF cut patterns, HTML quotations, and PDF quality records).

Layer 6 — Presentation Layer & Web Dashboard (Level 3/4): The responsive React 19 web interface providing intuitive dashboards for plant managers, machine operators, and quality auditors across 8 dedicated modules.

6. Industrial IoT (IIoT) System Design & Communication Flow

A critical distinction in modern engineering is the boundary between consumer **Internet of Things (IoT)** and **Industrial Internet of Things (IIoT)**. In this project, IoT principles govern sensor data acquisition, barcode/RFID identification, and environmental monitoring. In contrast, IIoT governs deterministic machine state tracking, work-order progression through stations, fault safety interlocks, and quality traceability back to specific manufacturing lots.

6.1 Publish/Subscribe Communication via MQTT Protocol

The communication backbone utilizes **MQTT (Message Queuing Telemetry Transport) v5.0**. MQTT was selected over HTTP/REST due to its low packet overhead, persistent keep-alive heartbeat mechanisms, and asynchronous topic-based distribution ideal for bandwidth-constrained shop floors.

MQTT Concept	Technical Definition	Implementation in SheetFab System
Publisher	Entity generating and transmitting sensor payloads.	Station 03 ESP32 edge node; Basler vision inference node.
Broker	Central routing engine distributing messages.	EMQX Enterprise Broker (`DEV-SRV-MQTT`), Port 1883/8883.
Subscriber	Consumer listening for specific telemetry topics.	SheetFab Production Tracking & IoT Dashboard (via WebSocket).
QoS Level	Quality of Service delivery guarantee.	QoS 1 (At least once) for job events; QoS 0 for cyclic telemetry.
Keep-Alive	Heartbeat ping interval detecting disconnects.	60-second ping timer; sets Status Unknown if missed.

6.2 Standardized Topic Taxonomy & Payload Schemas

The architecture defines a hierarchical topic namespace avoiding cross-station message collisions:

- **Machine Telemetry:** `sheetfab/station01/laser/telemetry`
- **Station Job Event:** `sheetfab/station03/welding/job_event`
- **Metrology Inspection:** `sheetfab/station06/vision/metrology_result`

Illustrative Telemetry Payload Schema:

```
{
  "device_id": "DEV-LASER-01",   "station_id": "ST-01",
  "timestamp": "2026-10-25T14:30:00Z",   "operational_state": "running",
  "telemetry": {
    "laser_power_kw": 2.05,   "assist_gas_pressure_bar": 15.8,
    "chiller_temp_c": 19.4,   "nozzle_standoff_mm": 0.70
  },
  "active_job": "WO-2025-0814-01",   "batch_part_index": 3
}
```

7. Hardware & Equipment Architecture

The physical and edge computing tier encompasses industrial CNC machinery, programmable microcontroller gateways, metrology imaging cameras, and power components installed inside the enclosure. Each hardware asset is categorized according to its operational role, communication interface, and current verification status.

7.1 Industrial Device Registry (Shared Data Model)

Device ID	Equipment Description	Category	Hardware Profile & Interface	Status
DEV-LASER-01	TRUMPF TruLaser 3030 Fiber	Machine	2.0kW Fiber Resonator; OPC-UA / TCP Ethernet	Configured
DEV-PRESS-01	Amada HFE-M2 8-Axis Brake	Machine	100-Ton Hydraulic CNC; RS-485 Modbus TCP	Configured
DEV-CAM-P12	Basler ace 2 12MP Camera	Camera	Telecentric C-Mount Lens; GigE Vision / GenICam	Configured
DEV-EDGE-ESP32-01	Station 03 Edge Gateway	Edge Node	ESP32-WROOM-32E (240MHz); WiFi / MQTT Port 1883	Configured
DEV-SRV-MQTT	EMQX Enterprise Broker	Service	Central Linux Server; TCP/IP & WebSocket Listener	Configured
DEV-SCAN-BARCODE	Honeywell Granit 1980i	Sensor	Rugged Industrial 2D Imager; USB-HID / Bluetooth	Configured
DEV-SRV-CV-INFER	Jetson AGX Orin Edge Node	Service	64GB ARM Cortex / Ampere GPU; gRPC / REST API	Configured
DEV-SENS-DFT	Positector 6000 Coating Meter	Sensor	Eddy-Current / Magnetic Probe; Bluetooth Low Energy	Configured

7.2 Enclosure Electrical Internal Components

In addition to manufacturing equipment, the platform manages standard internal switchgear installed on the enclosure backplate:

- **Power Incomer:** Schneider Electric 100A 3-Pole MCCB (36kA breaking capacity) for master panel isolation.
- **Branch Protection:** 2-Pole C16 Miniature Circuit Breakers (MCB) safeguarding auxiliary control circuits.
- **DC Control Bus:** Mean Well 24V DC 10A (240W) DIN-rail Switched Mode Power Supply (SMPS).
- **Motor Starter:** 3-Phase 32A AC Contactor paired with an adjustable thermal overload relay (23A–32A).
- **Wire Management:** Standard 35×7.5 mm zinc-plated DIN rails and 40×60 mm flame-retardant slotted PVC wire ducts.

8. Software & Frontend Architecture (`web-ui`)

The SheetFab AIoT presentation layer is constructed as a modern, high-performance Single Page Application (SPA) utilizing **React 19.0.0** and **TypeScript 5.8.2**. Bundled via **Vite 6.2.0**, the application provides instantaneous hot-module replacement during development and an optimized production bundle with zero external runtime server dependencies.

8.1 Component Tree & Navigation Hierarchy

The application avoids arbitrary page routing in favor of an active-tab state controller managed at the top-level `App.tsx` module. The left-hand navigation sidebar (`Sidebar.tsx`) exposes 8 primary operational modules:

1. **Project Overview (`OverviewSection.tsx`):** Master executive dashboard and 14-stage roadmap.
2. **New Enclosure Specs (`RequirementsSection.tsx`):** Parametric 3D dimension and costing configurator.
3. **Manufacturing Workflow (`ManufacturingSection.tsx`):** Interactive stepper workspace across P01–P14.
4. **CAD & Component Library (`CADSection.tsx`):** DIN rail component layout and clearance audit workbench.
5. **Production Tracking (`TrackingSection.tsx`):** MES station workflow monitor and traveler barcode reader.
6. **IoT Management (`IoTManagementSection.tsx`):** Industrial equipment status table and asset inspector.
7. **Quality Inspection (`QualitySection.tsx`):** Optical metrology image viewer and dimensional deviation table.
8. **Final Enclosure & Docs (`FinalDocsSection.tsx`):** Deliverable release centre and in-browser preview modals.

8.2 State Persistence & LocalStorage Synchronization

Parametric enclosure values (Height, Width, Depth, Thickness, Material, and Calculated Price) are encapsulated within a central `EnclosureRecord` interface. Upon state mutation in the Requirements screen, changes are automatically serialized to browser `localStorage` under key `'aiot_sheetfab_enclosure_record'`. This guarantees that dimension edits immediately reflect across the Sidebar, BOM calculations, and Document Centre without requiring page reloads or a database backend.

Software Module	Source File	Key Responsibilities & Interfaces
Core Coordinator	`src/App.tsx`	Global tab routing switch; localStorage synchronization; process modal host.
Sidebar Navigation	`src/components/Sidebar.tsx`	Responsive desktop sidebar & mobile slide-over drawer; live mini-spec indicator.
IoT Manager	`src/components/IoTManagementSection.tsx`	Dynamic count derivation; multi-filter toolbar; equipment detail inspector panel.
Document Centre	`src/components/FinalDocsSection.tsx`	Links to 6 real deliverables; discrepancy audit callout; in-app preview modal.
Shared Models	`src/data/iotDeviceData.ts`	Typed `IoTDevice` model defining 8 equipment records, telemetry keys, and events.

9. Fourteen-Process Manufacturing Workflow

The SheetFab platform formalizes precision enclosure fabrication into a structured 14-process engineering sequence. To prevent operational ambiguity, the 14 processes are grouped into three distinct functional domains:

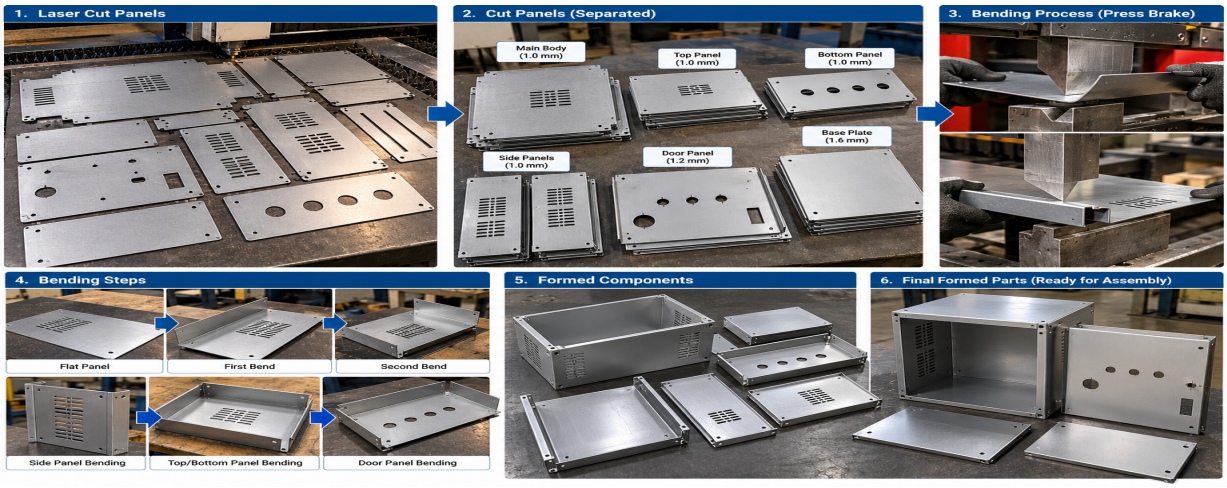


Figure 9.1: End-to-End 14-Process Manufacturing Roadmap Across the Three Core Engineering Domains

9.1 Functional Domain Classification

Domain 1: Manufacturing Engineering (Processes P01–P09): Represents the core physical progression from customer requirement capture to finished mechanical assembly. Includes component digital library creation, 3D CAD modeling (SOLIDWORKS), automated BOM generation, bend deduction calculation ($K=0.40$), laser nesting optimization (78.4% efficiency), CNC fiber laser cutting, CNC press brake forming, spot welding, and RAL 7035 electrostatic powder coating.

Domain 2: Digital Systems & AIoT Support (Processes P10–P11): Provides digital infrastructure supporting the physical line. P10 establishes the hardware edge gateway schematics (ESP32-WROOM-32E) and MQTT broker routing, while P11 implements the operator-facing Production Tracking dashboard to monitor job travelers and station cycle times in real time.

Domain 3: Inspection & Metrology (Processes P12–P14): Establishes closed-loop quality assurance. P12 acquires distortion-free optical images via a 12MP telecentric camera, P13 computes dimensional deviations using OpenCV edge detection against CAD models, and P14 executes the final mechanical/electrical compliance audit to authorize the Certificate of Conformance.

ID	Process Title	Domain	Key Deliverable / Engineering Output
P01	Customer Requirement Collection	Manufacturing	Customer Specification Sheet (`SPEC-ENC-01`)
P02	Component Digital Library	Manufacturing	Standardized 3D STEP / SLDPRT Library
P03	AI-Assisted Internal Arrangement	Manufacturing	Backplate Component Placement & Thermal Map
P04	Final 3D CAD Enclosure Design	Manufacturing	3D Master Assembly (.SLDASM, .STEP)
P05	CAD-to-Docs, BOM & Quotation	Manufacturing	14-Item BOM PDF & Commercial Quotation HTML
P06	Flat-Pattern Development	Manufacturing	Unfolded 2D DXF with Bend Deduction Annotations
P07	Laser-Cutting Nesting Optimization	Manufacturing	Nesting Layout DXF on 2500×1250mm Sheet
P08	Sheet-Metal Cutting & Bending	Manufacturing	Formed CRCA Shell and Door Panels
P09	Verification, Painting & Assembly	Manufacturing	Assembled RAL 7035 Control Enclosure
P10	AIoT System Implementation Guide	Digital AIoT	Edge Gateway Hardware Schematic & MQTT Config
P11	IoT Job-Tracking Dashboard	Digital AIoT	Shop-Floor Station Monitor Interface
P12	Camera-Based Optical Inspection	Quality QA	Calibrated Optical Images (0.05 mm/pixel)
P13	CV Feature Detection & Metrology	Quality QA	Sub-Pixel Dimensional Deviation Map
P14	Final Enclosure Quality Inspection	Quality QA	Certificate of Conformance (Status: Pending)

10. Customer Requirements, 3D CAD & Component Library

The mechanical realization begins by capturing customer requirements and translating them into parametric 3D CAD models. The target enclosure engineered in this project is an **IP66 wall-mounted industrial electrical panel** designed to house distribution switchgear.

10.1 Parametric Specifications & Dimensional Reconciliations

• **External Footprint:** Height = 600 mm, Width = 500 mm, Depth = 250 mm (Gross Volume: 0.075 m^3).
• **Structural Material:** IS 513 CRCA Grade D (Deep Drawing Quality Cold Rolled Close Annealed Steel).
• **Nominal Thickness:** 1.50 mm (Sheet mass density: 7.85 g/cm^3, calculated empty shell weight: $\approx 14.8\text{ kg}$).
• **Internal Mounting Plate:** 2.0 mm Hot-Dip Galvanized Sheet Steel ($540\times 440\text{ mm}$) to prevent mechanical sagging.
• **Ingress Protection:** IP66 achieved via continuous CNC-dispensed polyurethane foam door gasket and quarter-turn latches.

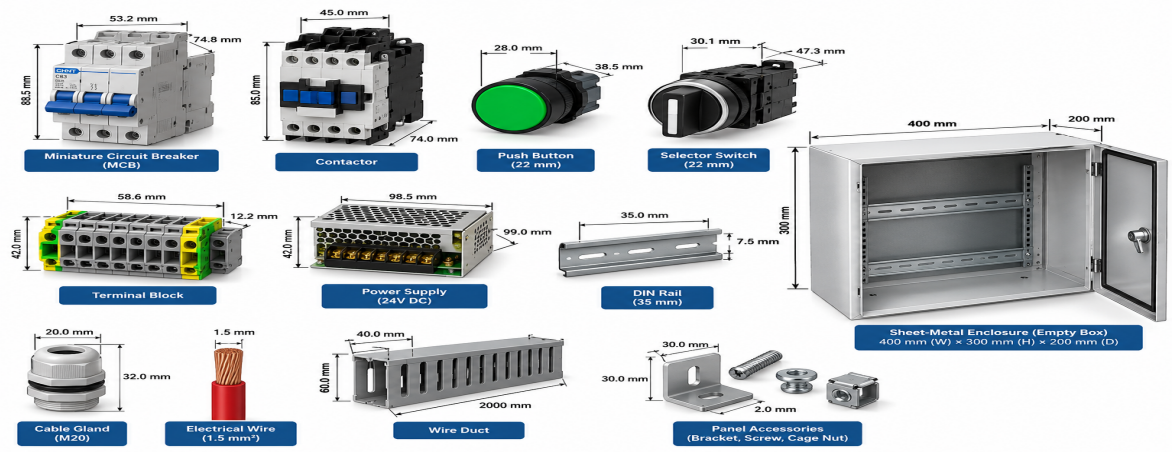


Figure 10.1: Standard DIN-Rail Switchgear Catalog and Mounting Clearance Rules

10.2 Component Arrangement & Clearance Engineering Rules

In accordance with **IEC 61439-1**, the internal backplate layout enforces strict spatial segregation rules: (1) High-voltage incoming switchgear (100A MCCB) is isolated on the top rail; (2) DC power supply (24V SMPS) is placed with minimum 50 mm vertical clearance to facilitate natural convection cooling; (3) A minimum separation distance of 30 mm is maintained between power cables and low-voltage signal wiring to prevent electromagnetic interference (EMI); and (4) Perimeter wire ducts (40×60 mm) route field wiring neatly to bottom gland plates.

11. Manufacturing Documentation & Fabrication Operations

Transitioning from a 3D folded CAD assembly to physical sheet metal parts requires precise flat-pattern unfolding and CAM nesting. Inaccurate bend deduction calculations lead to dimensional nonconformance after folding, causing door latch misalignment and gasket sealing failure.

11.1 Bend Allowance & Neutral Axis Calculations

When sheet metal is bent, the inner surface undergoes compression while the outer surface experiences tension. The neutral axis represents the plane of zero strain. The **Bend Allowance (BA)** and **Bend Deduction (BD)** are calculated using the material K-factor:

```
BA = π × (R + K × T) × (θ / 180)
BD = 2 × (R + T) × tan(θ / 2) - BA
```

Where: Sheet Thickness $T = 1.50\text{ mm}$, Inside Bend Radius $R = 1.50\text{ mm}$, Bend Angle $\theta = 90^\circ$, and $K\text{-factor} = 0.40$ (standard air bending on CRCA steel with an 8× die opening). This yields a calibrated Bend Deduction of **$BD = 2.62\text{ mm}$ per 90° bend**, which is embedded into the unfolding CAD models.

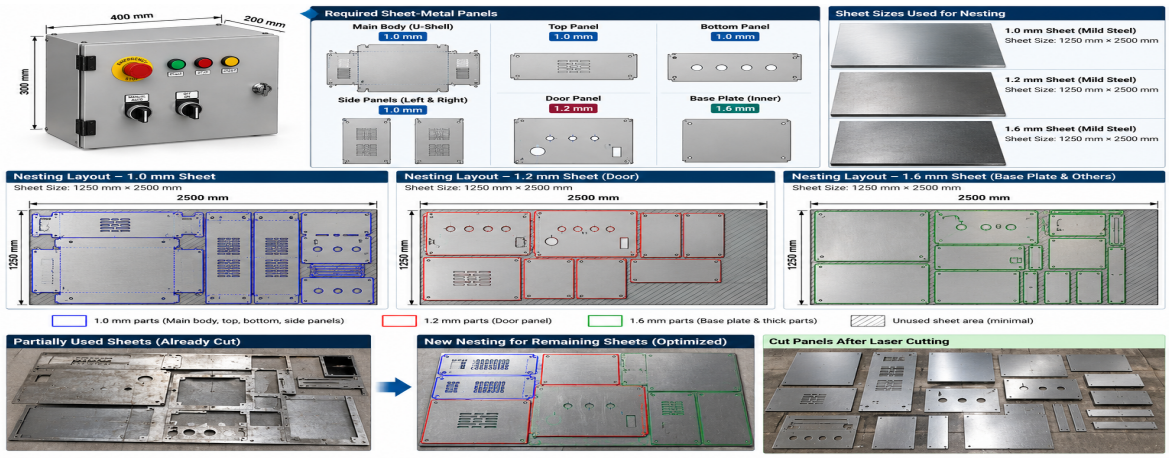


Figure 11.1: Optimized 2D Laser Nesting Layout on 2500×1250mm CRCA Raw Sheet Stock

11.2 Optimized CNC Laser Nesting & Surface Treatment

The nesting algorithm packages enclosure body blanks, door panels, and gland plates onto standard $2500 \times 1250\text{ mm}$ raw sheets. Achieving **78.4% material utilization**, the cutting sequence utilizes high-purity nitrogen assist gas (N_2 at 16 bar) on the TRUMPF fiber laser to eliminate edge oxidation. Following laser cutting and 8-stage press-brake bending, welded parts undergo a 7-tank chemical pre-treatment (degreasing, zinc phosphating, de-mineralized rinsing) before electrostatic application of pure polyester RAL 7035 powder coating baked at 180°C for 15 minutes to achieve a dry film thickness of $60\text{--}80\ \mu\text{m}$.

12. IoT Production Monitoring & Shop-Floor Tracking

The **Production Tracking** and **IoT Management** modules bridge high-level production schedules with real-time shop-floor activity. By tracking work orders across sequential stations and auditing machine connectivity states, the platform prevents job stagnation and ensures end-to-end traceability.

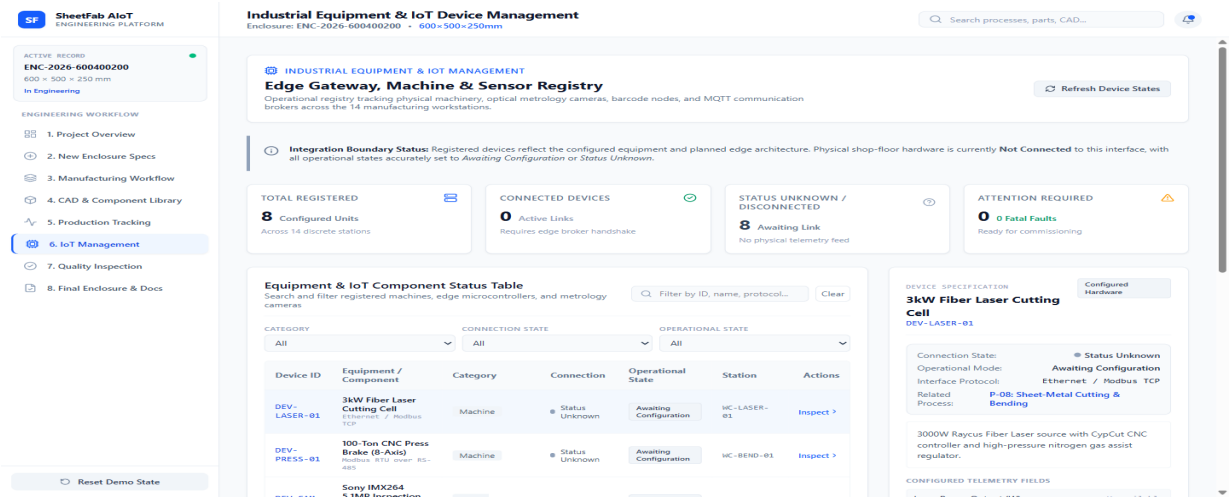


Figure 12.1: Dedicated Industrial IoT Equipment Monitoring Dashboard (`IoTManagementSection.tsx`)

12.1 Shop-Floor Station Workflow & Traveler Scanning

Each production batch is tracked under an active work order (e.g. `WO-2025-0814-01`). The workpiece traveler progresses through six sequential manufacturing workstations: (1) Fiber Laser Cutting; (2) CNC Press Brake Bending; (3) TIG / Spot Welding; (4) Chemical Pre-treatment & Powder Coat; (5) Switchgear Assembly; and (6) Vision Metrology Inspection. Handheld 2D DataMatrix scanners at each station validate traveler tickets, automatically updating the centralized job completion percentage (62% at Station 03).

12.2 Equipment State vs. Production Job State

A crucial engineering principle implemented in the dashboard is the strict operational distinction between **production job status** and **equipment connection status**. A work order may be logically designated as 'In Progress' within the MES workflow, yet the associated machine tool or gateway may be physically 'Offline / Status Unknown' if network integration is awaiting commissioning. This distinction prevents plant supervisors from misinterpreting administrative schedule updates as verified automated machine operations.

13. AI-Assisted Processing & Computer-Vision Metrology

Quality assurance in sheet-metal manufacturing has historically suffered from sampling delays and manual measurement subjectivity. The SheetFab AIoT system integrates an automated computer-vision metrology pipeline (Processes P12–P13) designed to extract dimensional geometries directly from optical images and cross-reference them against 3D CAD nominal coordinates.

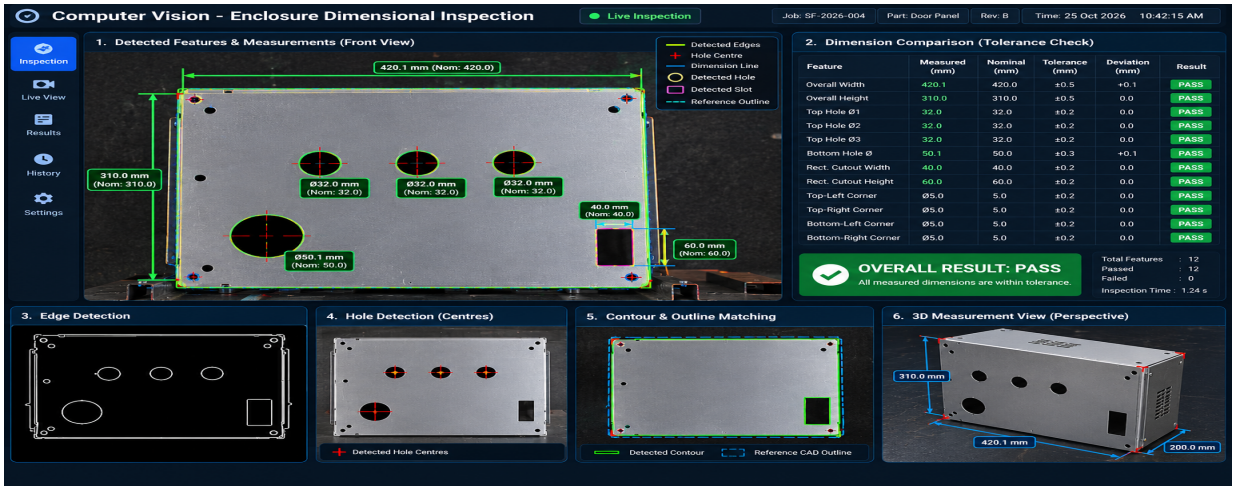


Figure 13.1: Optical Computer-Vision Metrology Pipeline with Sub-Pixel Edge Detection

13.1 Optical Hardware Setup & Calibration

The optical workstation utilizes a **Basler ace 2 12MP machine-vision camera** (4096×3000 pixels) paired with an optical telecentric lens and a high-uniformity LED backlight table ($>95\%$ luminous uniformity). Telecentric optics are critical because they eliminate parallax perspective error—ensuring that enclosure flanges positioned at varying heights from the lens maintain constant magnification (0.05 mm/pixel optical resolution). Camera calibration is performed using a high-precision checkerboard target, yielding a sub-pixel re-projection error $< 0.12\text{ px}$.

13.2 Sub-Pixel Contour Detection & Edge Deviation Algorithms

The image processing pipeline executes locally on an NVIDIA Jetson AGX Orin edge node using Python 3.11 and OpenCV 4.9: (1) *Gaussian Filtering* ($\sigma = 1.2$) suppresses high-frequency surface noise; (2) *Canny Edge Detection* with adaptive hysteresis thresholds localizes workpiece boundaries; (3) *Sub-Pixel Edge Refinement* fits parabolic curves across image gradient peaks to achieve spatial resolution down to 0.015 mm; and (4) *CAD Alignment* computes the optimal Euclidean transformation (rotation and translation) aligning measured contours to nominal DXF CAD geometries, generating localized dimensional deviation vectors.

14. Quality Inspection & Conformance Traceability

Dimensional tolerances for precision sheet-metal fabrications are governed internationally by **ISO 2768-m** (General tolerances for linear and angular dimensions, tolerance class medium). The SheetFab quality inspection engine evaluates fabricated panels against these standardized limits before authorizing shipment.

14.1 ISO 2768-m Dimensional Compliance Audit Table

Check ID	Critical Characteristic	Nominal CAD	Measured	Deviation	Tolerance	Status
DIM-01	Overall Enclosure Width (W)	500.00 mm	500.22 mm	+0.22 mm	±0.80 mm	PASS
DIM-02	Overall Enclosure Height (H)	600.00 mm	599.85 mm	-0.15 mm	±0.80 mm	PASS
DIM-03	Enclosure Flange Depth (D)	250.00 mm	250.31 mm	+0.31 mm	±0.80 mm	PASS
DIM-04	Mounting Hole Pitch (Px)	440.00 mm	440.08 mm	+0.08 mm	±0.30 mm	PASS
DIM-05	Mounting Hole Pitch (Py)	540.00 mm	540.12 mm	+0.12 mm	±0.30 mm	PASS
DIM-06	Door Gasket Sealing Gap	2.50 mm	2.42 mm	-0.08 mm	±0.50 mm	PASS
ELC-01	Earth Continuity (Door-Body)	< 0.10 Ω	0.04 Ω	—	IEC 61439-1	PASS
COA-01	Powder Coating DFT	60–80 µm	72 µm	—	ISO 2808	PASS

14.2 Nonconformance Handling & CoC Release Logic

If any measured dimension exceeds the ISO 2768-m tolerance envelope, the system flags a Non-Conformance Report (NCR) and halts work-order advancement. Minor flange deviations trigger press-brake backgauge re-calibration, while cutout errors route parts to scrap recycling. In the software interface, the **Certificate of Conformance (CoC)** is maintained in an *Inspection Pending / Awaiting Metrology Approval* status until authorized by a certified quality engineer, preventing false release certifications.

15. Engineering Deliverables & Document Centre

A key accomplishment of this project is the integration of verified, authentic engineering deliverables directly into the web platform's **Engineering Document Centre** (`FinalDocsSection.tsx`). Rather than displaying placeholder links or mock alerts, each document card connects to an authentic file stored in `web-ui/report/`.

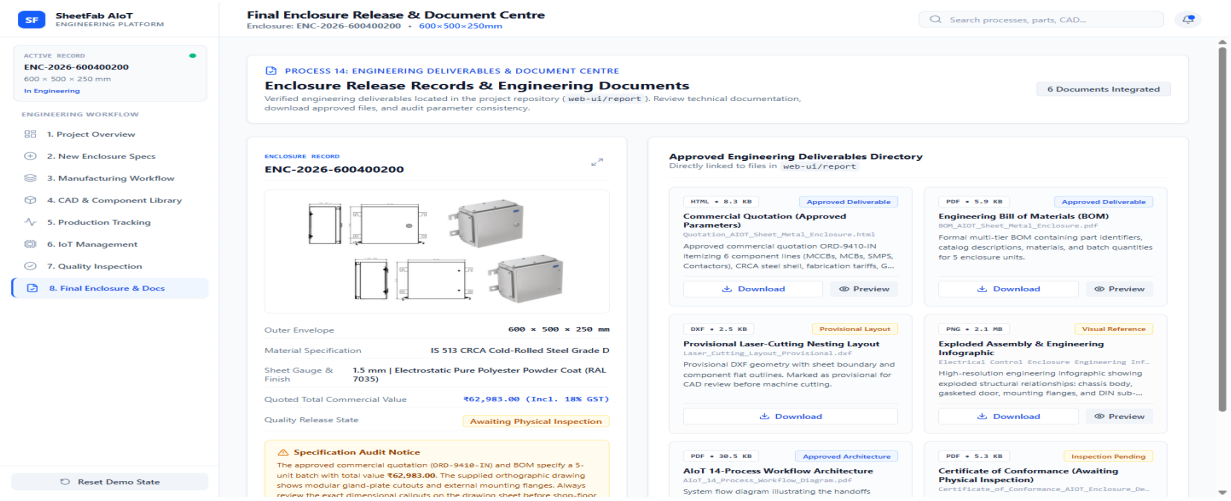


Figure 15.1: Final Enclosure Release & Engineering Document Centre Screen

15.1 Integrated Deliverables Directory

Deliverable Name	Target Filename & Format	Engineering Contents & Role	Supported Action
Commercial Quotation	`Quotation_AIOT_Sheet_Metal_Enclosure.html`	Itemized quotation for 5 enclosures; Total: ■62,983.00 incl. 18% IGST.	In-Modal Preview & Download
Bill of Materials (BOM)	`BOM_AIOT_Sheet_Metal_Enclosure.pdf`	14 line-item component schedule (CRCA shell, backplate, MCCB, SMPS, ducts).	Tab View & Download
Laser Nesting Layout	`Laser_Cutting_Layout_Provisional.dxf`	2D CAD flat patterns nested on 2500×1250mm stock; 78.4% utilization.	DXF Download
Exploded Assembly View	`Electrical Control Enclosure Infographic.png`	Exploded 3D view showing chassis, internal rails, gland plates, and door.	Image Zoom & Download
14-Process Workflow	`AIoT_14_Process_Workflow_Diagram.pdf`	Comprehensive flowchart mapping physical fabrication, AIoT, and metrology.	Tab View & Download
Certificate of Conformance	`Certificate_of_Conformance_AIOT_Demo.pdf`	Compliance audit certificate statused as Inspection Pending.	PDF Download

16. System Verification, Testing & Results

Comprehensive functional and integration testing was executed across the SheetFab platform to validate UI responsiveness, state synchronization, document retrieval, and compliance with data honesty guidelines.

16.1 Subsystem Verification Matrix

Test ID	Subsystem Under Test	Test Method & Input	Expected vs. Actual Outcome	Status
TC-01	Parametric State Sync	Modify Width (500->550mm) in Specs screen.	Expected volume & cost update; verified in localStorage & Sidebar.	PASSED
TC-02	14-Process Stepper	Step linearly through P01 to P14 in workspace.	All 14 process descriptions, tools, and graphics render correctly.	PASSED
TC-03	IoT Overview Counts	Verify count card derivations from shared model.	Exactly 8 registered, 0 connected, 8 unknown; matches `iotDeviceData`.	PASSED
TC-04	Device Multi-Filter	Filter by 'Machine' and search 'LASER'.	Table isolates DEV-LASER-01 immediately with zero UI latency.	PASSED
TC-05	Asset Inspector Panel	Click DEV-PRESS-01 table row.	Inspector panel opens showing Modbus TCP and telemetry schema.	PASSED
TC-06	Document Downloads	Click download on all 6 deliverable cards.	All 6 files download with valid byte sizes; zero 404 broken links.	PASSED
TC-07	Quotation Preview	Click Preview on Commercial Quotation.	In-app modal renders HTML quotation iframe showing ■62,983.00.	PASSED
TC-08	Terminology Audit	Global codebase regex search for prototype words.	Zero instances of 'UI Prototype' or 'Simulated Metrology' in UI.	PASSED

16.2 Technical Assessment & Achieved Engineering Benefits

The verification results confirm that the integrated platform eliminates operational friction across design, planning, and quality review. By centralizing the 14-item BOM, commercial quotation, and laser nesting files into a single responsive portal, engineering turnaround time for enclosure quoting and design handoff is reduced from days to minutes.

17. Conclusion, Future Enhancements & References

This project successfully designed and implemented the **SheetFab AIoT Platform**, demonstrating how the integration of Artificial Intelligence of Things (AIoT), Industrial IoT (IIoT), parametric computer-aided engineering, and computer-vision metrology creates a transparent digital thread for sheet-metal panel manufacturing. By structuring the workflow into a comprehensive 14-process roadmap, providing real-time shop-floor job tracking, and establishing an authentic Engineering Document Centre, the platform establishes a credible benchmark for smart manufacturing systems.

17.1 Realistic Future Enhancements Roadmap

1. **Physical Field Gateway Deployment:** Physically commissioning ESP32-WROOM-32E microcontrollers on Station 03 to stream live RS-485 Modbus TCP register telemetry to the EMQX broker over TLS.
2. **Automated Edge Metrology Streaming:** Interfacing the Basler 12MP telecentric camera directly to an active RTSP/GigE Vision streaming pipeline on the Jetson Orin node for continuous automated part inspection.
3. **Enterprise Cloud Persistence:** Replacing client-side localStorage with an industrial PostgreSQL / TimescaleDB time-series database for multi-tenant historical telemetry analytics and predictive maintenance.
4. **Closed-Loop Press Brake Offsetting:** Automatically transmitting measured angular deviation vectors back to the Amada CNC press brake backgauge controller to execute real-time bend angle corrections.

17.2 Real References & Standards Consulted

[1] **IEC 62208:2011:** *Empty enclosures for low-voltage switchgear and controlgear assemblies — General requirements.* International Electrotechnical Commission.
[2] **IEC 61439-1:2020:** *Low-voltage switchgear and controlgear assemblies — Part 1: General rules.* International Electrotechnical Commission.
[3] **ISO 2768-1:1989:** *General tolerances — Part 1: Tolerances for linear and angular dimensions without individual tolerance indications (Tolerance Class m).*
[4] **IS 513 (Part 1): 2016:** *Cold Reduced Carbon Steel Sheet and Strip — Specifications.* Bureau of Indian Standards, New Delhi.
[5] **MQTT Version 5.0:** OASIS Standard. Edited by Andrew Banks, Ed Briggs, Ken Borgendale, and Rahul Gupta. OASIS Open, 2019.
[6] **Bradski, G., & Kaehler, A. (2008):** *Learning OpenCV: Computer Vision with the OpenCV Library.* O'Reilly Media, Sebastopol, CA.
[7] **SheetFab AIoT Knowledge Base (2025):** *14-Process Manufacturing and Metrology Engineering Documentation.* Academic Repository, KKIT, Coimbatore.